Cardiomyopathies: An Approach to a Case of Dyspnoea

Dr. Muhammad Reihan, Internal Medicine | Source: `Reihan_Cardiomyopathies.pdf` (29 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The case answer is **reasoned by me; the key is not on the slides**.

🔗 **Bridge from earlier lectures:**

- **Heart Failure (same lecturer):** DCM = **HFrEF** (all GDMT applies); HCM and RCM = **HFpEF** physiology; the **ICD criteria** (EF ≤ 35%, NYHA II–III, 3 months GDMT) come from there.
- **Valvular heart disease:** the **HOCM murmur vs aortic stenosis**; both crescendo–decrescendo; **SAM → MR**.
- **Pericardial disease: restrictive cardiomyopathy vs constrictive pericarditis**, the classic look-alike.
- **Arrhythmias: tachycardia-induced cardiomyopathy** (AF); HCM → AF and **VT/SCD**.
- **Substance use / psychotropics: alcohol**, **cocaine**, **beriberi (thiamine)**; *[added]* **clozapine myocarditis**.
- **Endocrine:** thyroid disease and **acromegaly** (secondary DCM); **haemochromatosis** (RCM).
- **Restrictive lung / sarcoid / amyloid:** infiltrative causes overlap.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	⭐ Aetiology of **dilated, hypertrophic, restrictive** cardiomyopathy	ABCCCDD; PLEASe Help
Knowledge	⭐ Pathophysiology of each	Eccentric vs concentric hypertrophy; SAM
Skill	⭐ Features that warrant an **ICD** for sudden-death risk	HCM risk factors; EF ≤ 35%
Skill	Evidence-based management of each type	β-blockers in HCM; avoid nitrates/diuretics

Main subtopics:

1. overview and classification
2. **DCM** (epidemiology, ABCCCDD, pathophysiology, features, diagnostics, management)
3. **HCM** (epidemiology, genetics, obstructive vs non-obstructive, features, examination, diagnostics, management)
4. **RCM** (PLEASe Help, pathophysiology, features, diagnostics, management)
5. ⭐ **echo differentiation** (LV cavity, EF, wall thickness)

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2. Main Content (Lecture Order)

2.0 Opening Case (Slides 4–5, 26)

Presentation:

- **48-year-old woman**, 2 months of progressive exertional dyspnoea and fatigue
- no illness or medication; **father died of amyloidosis**; no smoking or alcohol
- **HR 124**, BP 98/60, no murmurs, **bibasal coarse crackles**
- **ECG: irregularly irregular, absent P waves (AF)**
- **CXR:** globular cardiomegaly, prominent hila, bilateral fluffy infiltrates (pulmonary oedema)
- **echo: DILATED LV, EF 40%**

Most likely cause? → ⭐ **d. Chronic tachycardia** (tachycardia-induced cardiomyopathy) *[reasoning added; key not on slides]*:

- **DCM** (dilated LV, ↓ EF) in a patient with **persistent fast AF** and **no other cause** (no alcohol, no CAD risk profile, no thiamine deficiency) fits **tachycardia-mediated DCM**
- slide 9 lists "**chronic tachycardia**" as a secondary cause of DCM
- *[added]* it is **reversible** with rate or rhythm control

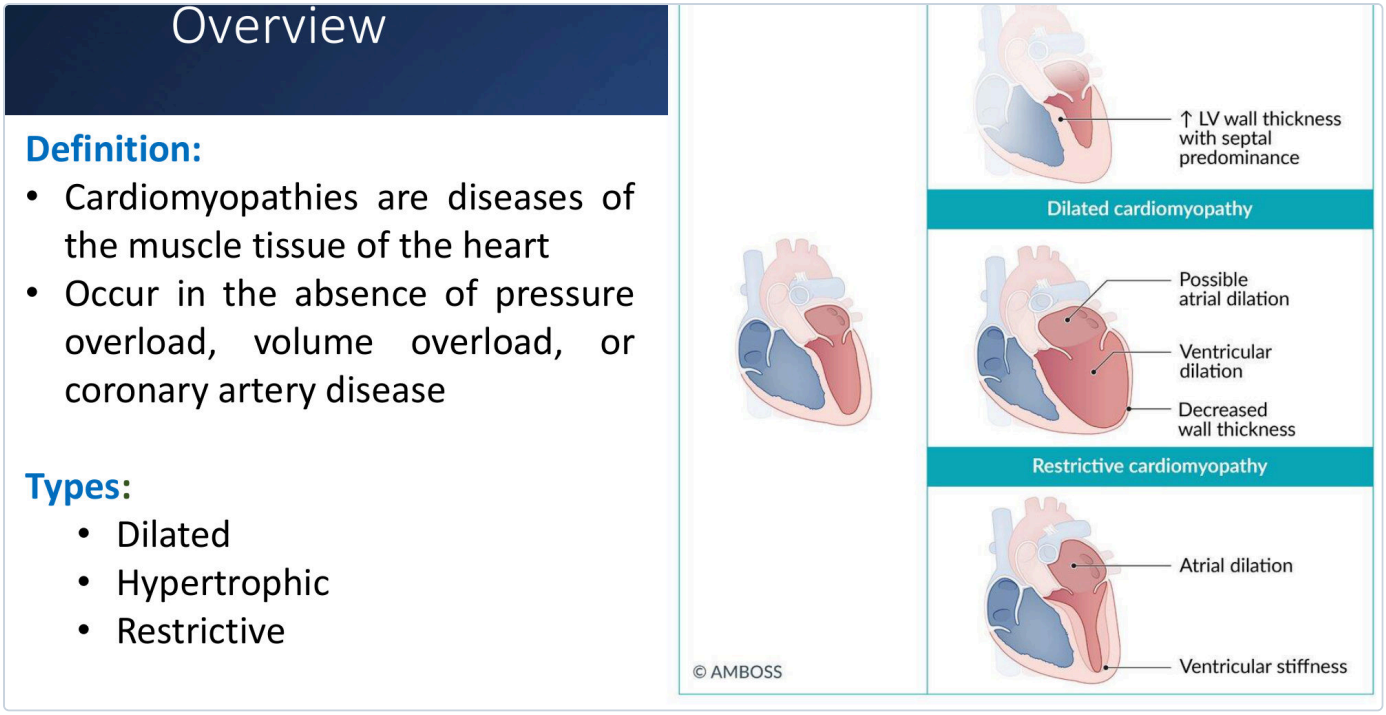
Why not the others:

- **a. amyloid** (the distractor via the father's history): **restrictive** CM with a **non-dilated, thick-walled** LV (and low-voltage ECG)
- **b. stress (takotsubo):** acute apical ballooning after a trigger
- **c. thiamine:** needs malnutrition or alcohol
- **e. coronary occlusion:** ischaemic history, regional wall-motion abnormality

2.1 Overview and Classification (Slide 6) ⭐

- ⭐ **Cardiomyopathies are diseases of the heart muscle itself**, occurring **in the absence of pressure overload, volume overload, or CAD**.
- **Types: dilated · hypertrophic · restrictive** (*[added]* plus **ARVC** and unclassified, e.g. takotsubo).

Type	Morphology (slide 6/25)
HCM	⭐ **↑ LV wall thickness, septal predominance**
DCM	**Ventricular dilation**, **↓ wall thickness**, possible atrial dilation
RCM	⭐ **Atrial dilation**, **ventricular stiffness** (normal-sized ventricles)



Slide 6: Cardiomyopathy types

2.2 Dilated Cardiomyopathy (Slides 7–12) ⭐⭐

Epidemiology:

- incidence **6/100,000/year**
- ⭐ **the most common cardiomyopathy**
- ⭐ **M:F ≈ 3:1**

Definition: ⭐ **↓ contractile function** of the ventricles → **cardiac enlargement and systolic dysfunction**.

Aetiology of HF / DCM (slide 8, Felker NEJM 2000):

- **Causes of HF: ischaemic 40%, DCM 37%**, valvular 12%, HTN 11%.
- **Within DCM:**

Cause	Share
⭐ **Idiopathic**	**50%** (of which ⭐ **~40% familial**)
Myocarditis	10%
Infiltrative	5%
Toxins (**EtOH, cocaine**)	4%
Peripartum	4%
HIV	4%
Connective tissue disease	3%
Chemotherapy (adriamycin)	1%
Other (tachycardia, congenital HD, stress-related, **Chagas**)	17%

Secondary causes (slide 9):

- ⭐ **peripartum cardiomyopathy**
- ⭐ **infection: Coxsackie B, Chagas disease**

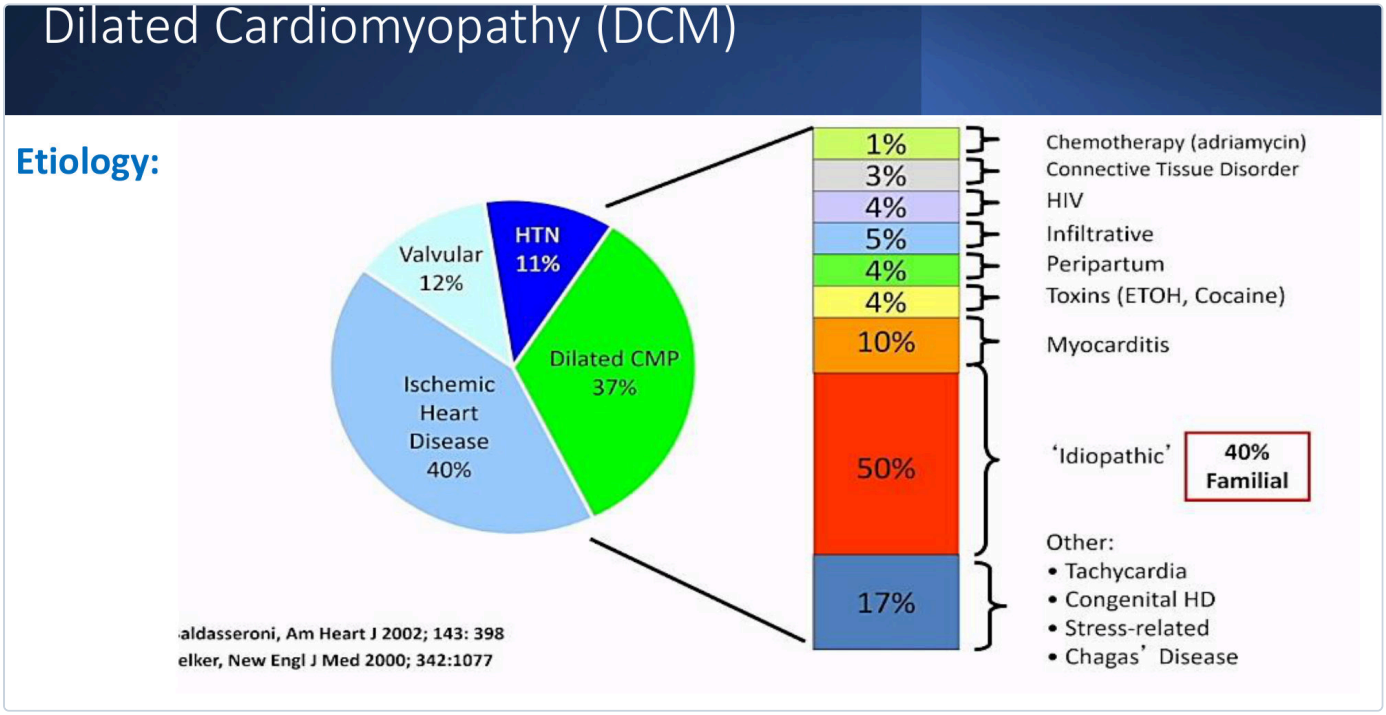
- **sarcoidosis**
- **radiation and chemotherapy**
- toxic substances
- **malnutrition (thiamine deficiency)**
- ⭐ **chronic (prolonged) tachycardia** (slide lists it twice)
- **endocrinopathies**

⭐ **Mnemonic ABCCCDD:**

- **A**lcohol
- **B**eriberi (thiamine)
- **C**ocaine
- **C**oxsackie B
- **C**hagas
- **D**oxorubicin/**D**aunorubicin

[added]

- **Anthracycline cardiotoxicity** is **dose-dependent** (cumulative doxorubicin > 450–550 mg/m²), so monitor EF.
- **Trastuzumab** is **reversible** (dose-independent).
- **Peripartum CM** = last month of pregnancy to 5 months postpartum.
- **Alcoholic CM is reversible** with abstinence.



Slide 8: Aetiology of HF/DCM

Pathophysiology (slide 10):

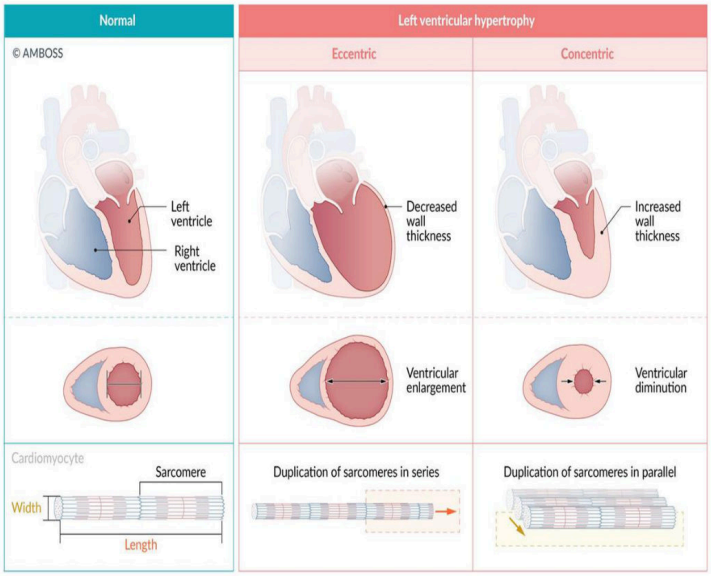
1. causative factors **↓ contractility**
2. ⭐ **ECCENTRIC hypertrophy: sarcomeres added in SERIES**
3. **ventricular dilation**
4. **systolic dysfunction, ↓ EF**
5. **heart failure**

[added] Dilation also stretches the mitral annulus → **functional MR**.

Dilated Cardiomyopathy (DCM)

Pathophysiology:

- Causative factors ↓ the contractility of the myocardium
- **Eccentric hypertrophy** (Sarcomeres added in series)
- Dilation of the ventricle
- Systolic dysfunction and decreased ejection fraction
- Heart failure



Slide 10: Eccentric vs concentric hypertrophy

Symptoms	Signs
⭐ **Exertional dyspnoea**	⭐ **S3 gallop**
Ankle oedema, ascites	⭐ **Displaced LV impulse**
Angina	**JVD**
Palpitations	**Bilateral rales**
—	**Diffuse abdominal and peripheral oedema**

Diagnostics (slide 12):

- **↑ BNP**
- **troponin and CK-MB**
- ⭐ **echocardiography**
- **CXR** (signs of HF)
- **ECG** (nonspecific)
- *[added]* **coronary angiography/CTCA to exclude ischaemic cause**; cardiac MRI (myocarditis, sarcoid); genetic testing if familial

DCM echo	
LV cavity	⭐ **Significantly ↑**
EF	⭐ **Significantly ↓** (systolic dysfunction)
Wall thickness	**Normal or ↓**

Management: ⭐ **treat the underlying disease** + ⭐ **treat HF** (full HFrEF GDMT, ICD/CRT as per the HF lecture).

- *[added]* **Anticoagulate** if AF or LV thrombus.

Dilated Cardiomyopathy (DCM)

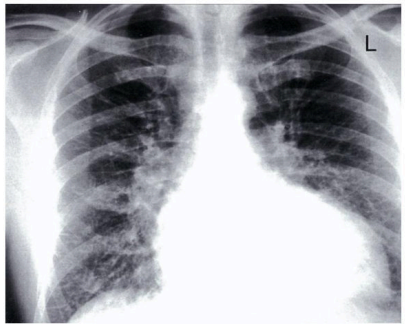
Diagnostics:

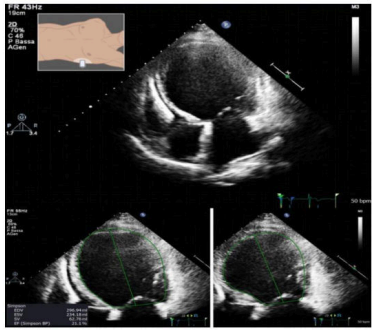
- ↑ BNP
- Troponin and CK-MB
- **Echocardiography**
- Chest x-ray (signs of HF)
- ECG (Nonspecific findings)

Management:

- Treatment of the underlying disease
- Treatment of heart failure

Echocardiography		
LV Cavity Size	**EF**	**Wall Thickness**
Significantly increased	Significantly decreased (Systolic dysfunction)	Normal or decreased





Slide 12: DCM diagnostics and echo

🔁 **Recap: DCM.**

- Commonest CM (6/100,000; M 3:1); ↓ contractility → eccentric hypertrophy (series) → dilated, thin LV → ↓ EF.
- Idiopathic 50% (40% familial).
- **ABCCCDD**; also peripartum, sarcoid, radiation, **chronic tachycardia**, endocrine.
- S3, displaced apex, HF signs.
- Echo: big cavity, low EF, thin/normal walls → **treat the cause + HFrEF GDMT**.

2.3 Hypertrophic Cardiomyopathy (Slides 13–19) ⭐⭐

Definition:

- ⭐ **LV hypertrophy with non-dilated chambers without an apparent cause** (no HTN or AS)
- ⭐ **genetic, autosomal dominant**

Epidemiology:

- ⭐ **1 in 500** people
- ⭐ **one of the most frequent causes of sudden cardiac death in the young, especially athletes**

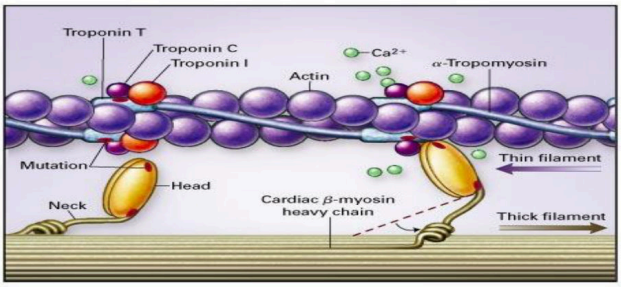
Genetics (slide 14):

- **autosomal dominant**: 50% of children of an affected parent
- **sarcomere mutations:** ⭐ **myosin-binding protein C** and ⭐ **β-myosin heavy chain**
- some cases are **spontaneous mutations**

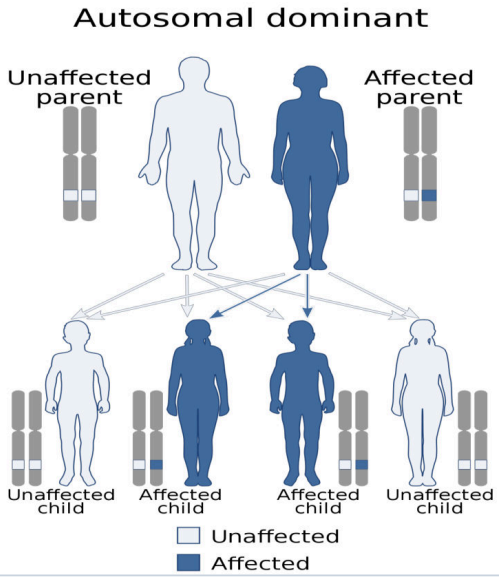
Hypertrophic Cardiomyopathy (HCM)

Etiology:

- Most cases are inherited as an autosomal-dominant trait
  - Myosin binding protein C
  - β-Myosin heavy chain
- Spontaneous mutations may account for some cases



N Engl J Med 2000; 343:1688-1696



Slide 14: HCM genetics (AD; sarcomere proteins)

Pathophysiology (slide 15):

1. ⭐ **CONCENTRIC hypertrophy: sarcomeres added in PARALLEL**
2. ⭐ **↓ ventricular relaxation** → **↓ diastolic filling** and ↓ systolic output (diastolic dysfunction)
3. ⭐ **Obstructive type (HOCM): LVOT obstruction** (asymmetric septal hypertrophy) + **mitral regurgitation** from ⭐ **systolic anterior motion (SAM)** of the mitral valve

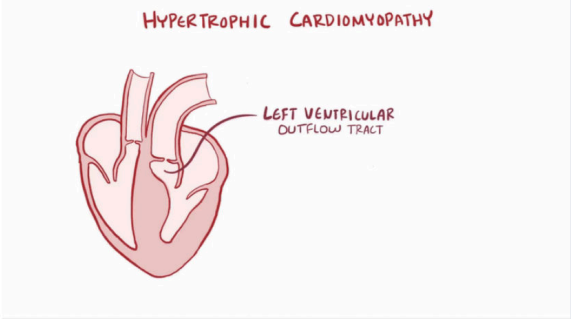
[added]

- **Non-obstructive vs obstructive:** about ⅓ obstructed at rest, ⅓ only on provocation, ⅓ never.
- Ischaemia comes from thick muscle with high O_2 demand and compressed small vessels.

Hypertrophic Cardiomyopathy (HCM)

Pathophysiology:

- **Concentric hypertrophy** of the left ventricle (Sarcomeres added in parallel)
- ↓ ventricular relaxation
- ↓ diastolic filling and systolic output
- Obstructive type (HOCM): left ventricular outflow tract obstruction & mitral regurgitation due to systolic anterior movement of the mitral valve



Slide 15: HCM/HOCM pathophysiology

Symptoms (slide 16):

- ⭐ **many are asymptomatic**
- **HF symptoms**
- **angina**
- ⭐ **syncope/presyncope** (*[added]* exertional)
- **palpitations, arrhythmias** (*[added]* AF, VT)
- ⭐ **sudden cardiac death, particularly during or after intense exercise**

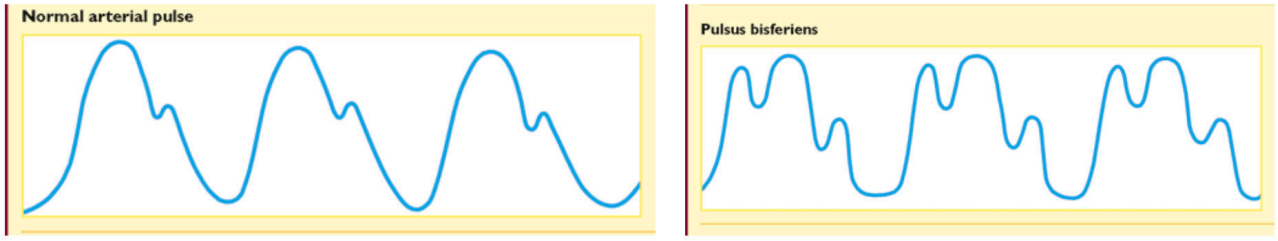
Examination (slide 17):

- ⭐ **systolic ejection (crescendo-decrescendo) murmur** (*[added]* left sternal edge, **no carotid radiation**; **↑ with Valsalva/standing, ↓ with squatting/handgrip**)
- ⭐ **S4 gallop**
- ⭐ **pulsus bisferiens:** LVOT obstruction gives a **quick rise of the pulse, then a slower, longer rise** (double-peaked)
- *[added]* "**jerky**" pulse; **double/triple apical impulse**

Hypertrophic Cardiomyopathy (HCM)

Physical examination:

- Systolic ejection murmur (crescendo-decrescendo)
- S4 gallop
- Pulsus bisferiens:
  LV outflow obstruction causes a sudden quick rise of the pulse followed by a slower longer rise



Slide 17: Pulsus bisferiens

HOCM murmur: smaller LV cavity = louder (opposite of aortic stenosis)

Rule: anything that ↓ preload or ↓ afterload shrinks the LV → septum and mitral leaflet (SAM) meet → more LVOT obstruction

Manoeuvre	Effect on LV volume	HOCM murmur	AS murmur
Valsalva (strain phase)	↓ preload	**↑ LOUDER**	↓ softer
Standing from squatting	↓ preload	**↑ LOUDER**	↓ softer
Nitrates / amyl nitrite, dehydration	↓ preload / ↓ afterload	**↑ LOUDER**	↑ (↓ afterload)
Squatting / passive leg raise	↑ preload (+ ↑ afterload)	**↓ softer**	↑ louder
Handgrip	↑ afterload	**↓ softer**	↓ softer
β-blocker (slows HR, ↑ filling)	↑ filling time	**↓ softer**	—

HOCM murmur: harsh systolic crescendo–decrescendo at the left sternal edge, NO carotid radiation; + MR murmur (SAM)
Avoid in HOCM: nitrates, diuretics (↓ preload), digoxin/inotropes, vasodilators (dihydropyridines)
Treat: β-blocker → verapamil → disopyramide/mavacamten → myectomy or alcohol ablation; ICD if high risk
Hypotension in HOCM: fluids + phenylephrine (pure α), NOT inotropes · whole table [added]

Added diagram: HOCM murmur manoeuvres vs aortic stenosis

Diagnostics (slide 18):

- **ECG:** septal/LVH (*[added]* deep narrow "dagger" Q waves in inferolateral leads; **apical HCM → giant T-wave inversion**)
- ⭐ **echo: wall thickness ≥ 15 mm**
- **MRI** case by case (*[added]* LGE fibrosis = SCD risk)
- ⭐ **genetic testing** to confirm and **screen family members**

HCM echo	
LV cavity	⭐ ↓
EF	**Normal** (diastolic dysfunction; *[added]* often **hyperdynamic > 70%**)
Wall thickness	⭐ **Significantly ↑** (≥ 15 mm; *[added]* ≥ 13 mm in relatives)

Management (slide 19):

- ⭐ **avoid strenuous exercise** (competitive sports)
- ⭐ **β-blockers** (first line)
- ⭐ **non-DHP CCB (verapamil)**
- **surgery for severe disease** (*[added]* **septal myectomy** or **alcohol septal ablation** for refractory LVOT gradient ≥ 50 mmHg)
- ⭐ **ICD for SCD prevention, guided by risk factors**
- *[added]* **mavacamten** (cardiac myosin inhibitor) for symptomatic obstructive HCM
- *[added]* **AVOID: nitrates, diuretics (high dose), digoxin, vasodilators, inotropes** (they worsen the obstruction)
- *[added]* **AF in HCM → anticoagulate regardless of CHA_2DS_2-VASc**

[added] **HCM SCD risk factors (ICD indications):**

1. **prior cardiac arrest / sustained VT** (secondary prevention)
2. **family history of SCD**
3. **unexplained syncope**
4. **massive LVH ≥ 30 mm**
5. **NSVT on Holter**

6. **LV apical aneurysm**
7. **EF < 50%**
8. **extensive LGE on MRI**

Hypertrophic Cardiomyopathy (HCM)

Management:

- Avoidance of strenuous exercise
- Beta blockers
- Non-dihydropyridine CCB
- Surgery generally preserved for severe disease
- ICD therapy for prevention of sudden death (guided by risk factors)

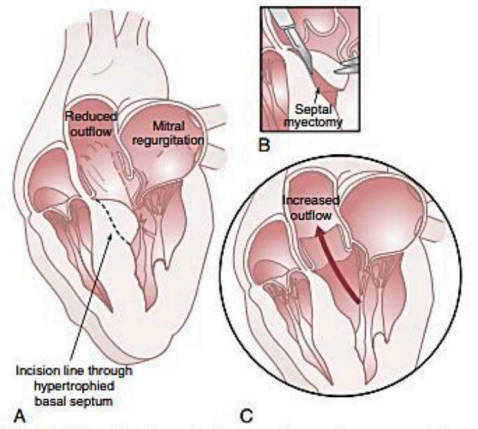


Fig. 10.3 (A to C) Schematic diagrams of a septal myectomy. (From Nishimura RA, Holmes DR Jr: Clinical practice: hypertrophic obstructive cardiomyopathy, N Engl J Med 350:1320-1327, 2004.)

Slide 19: HCM management

🔁 **Recap: HCM.**

- AD (MYBPC3, MYH7), 1/500; SCD in young athletes.
- Concentric hypertrophy (parallel) → diastolic dysfunction; HOCM = LVOT obstruction + **SAM → MR**.
- Syncope, angina, HF, SCD on exertion.
- Systolic murmur (↑ Valsalva, ↓ squat), **S4, pulsus bisferiens**.
- Echo: wall ≥ 15 mm, small cavity, normal EF; genetic family screening.
- **β-blocker / verapamil**, no strenuous exercise, myectomy, **ICD by risk**; avoid preload reducers.

2.4 Restrictive Cardiomyopathy (Slides 20–24) ⭐

Definition: ⭐ impaired ventricular filling of NON-dilated ventricles; genetic or acquired.

Epidemiology:

- ⭐ **rarest: 2–5%** of cardiomyopathies
- **HF in > 80%** of patients

Aetiology (slide 21):

- ⭐ **amyloidosis** (*[added]* commonest in adults; AL or transthyretin)
- ⭐ **sarcoidosis**
- ⭐ **hereditary haemochromatosis**
- **hypereosinophilia (Löffler endocarditis)**
- **endocardial fibroelastosis**
- **radiotherapy**

- **cardiac surgery**

⭐ **Mnemonic "PLEASe Help":**

- **P**ost-radiation / **P**ost-surgery fibrosis
- **L**öffler endocarditis
- **E**ndocardial fibroelastosis
- **A**myloidosis
- **S**arcoidosis
- **H**aemochromatosis

Pathophysiology (slide 22):

1. ⭐ **infiltration** (abnormal proteins, glycogen, eosinophils, granulomas, **iron**) or **fibrosis**
2. **↓ myocardial elasticity → ↓ compliance → ↓ diastolic filling**
3. ⭐ **↑ left- and right-sided filling pressures** → ⭐ **↑ atrial size (biatrial enlargement)**
4. **pulmonary AND systemic venous congestion**

Clinical features (slide 23):

- **pulmonary and systemic congestion**
- dyspnoea, palpitations, fatigue, weakness
- ⭐ **hepatosplenomegaly, ascites** (right-sided signs prominent)
- **signs of the underlying disease**
- *[added]* **Kussmaul's sign**, **S3** may be present, prominent **JVP** with rapid y-descent
- *[added]* **amyloid clues: low-voltage ECG despite thick walls**, macroglossia, periorbital purpura, carpal tunnel, nephrotic syndrome, "**apple-green birefringence** on Congo red"

Diagnostics (slide 24):

- **echocardiography**
- ⭐ **cardiac MRI** (*[added]* amyloid: diffuse subendocardial LGE)
- **CXR** (pulmonary congestion)
- ⭐ **endomyocardial biopsy** (definitive tissue diagnosis)

RCM echo (slide)	
LV cavity	**Typically normal**
EF	**Normal** (diastolic dysfunction)
Wall thickness	**Typically normal**

⚠️ **Caveat:** in **infiltrative RCM (amyloid, Fabry, haemochromatosis) the walls ARE thickened** ("pseudo-hypertrophy"), with **biatrial enlargement** and a **granular "sparkling" myocardium**. "Typically normal" applies mainly to fibrotic causes.

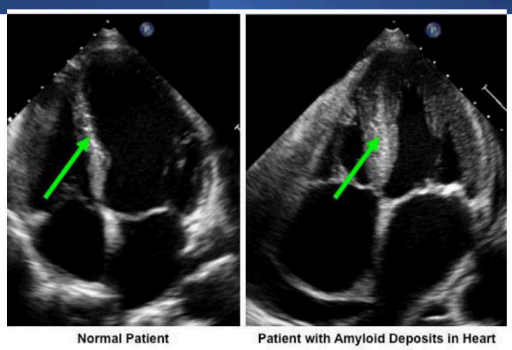
Management:

- ⭐ **treat the underlying cause**
  - *[added]* amyloid: **tafamidis** for transthyretin; chemo for AL
  - haemochromatosis: **phlebotomy**/chelation
  - sarcoid: steroids
- **manage HF** (cautious diuretics; *[added]* **avoid digoxin in amyloid**: toxicity)

Restrictive Cardiomyopathy (RCM)

Diagnostics:

- Echocardiography
- Cardiac MRI (CMR)
- CXR (Signs of pulmonary congestion)
- Endomyocardial Biopsy



Management:

- Treat the underlying cause
- Management of HF

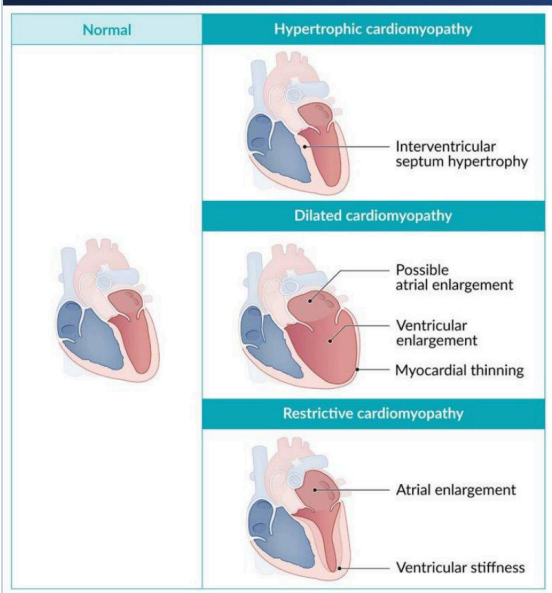
Echocardiography		
LV Cavity Size	**EF**	**Wall Thickness**
Typically normal	Normal (Diastolic dysfunction)	Typically normal

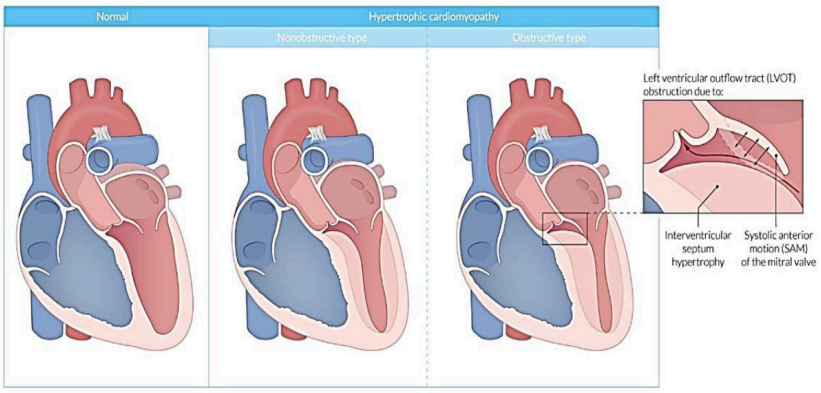
Slide 24: RCM diagnostics and echo

2.5 Summary Diagram (Slide 25)

- Normal vs **HCM** (septal hypertrophy) vs **DCM** (atrial and ventricular enlargement, myocardial thinning) vs **RCM** (atrial enlargement, ventricular stiffness).
- **Obstructive HCM:** LVOT obstruction from **septal hypertrophy + SAM**.

To Summarize





Slide 25: Summary, normal vs HCM vs DCM vs RCM

🔁 **Recap: RCM.** Rarest (2–5%); non-dilated, stiff ventricles; ↑ filling pressures → **biatrial enlargement**, systemic > pulmonary congestion. **PLEASe Help** (amyloid, sarcoid, haemochromatosis, Löffler, EFE, post-radiation/surgery). Echo/MRI/**endomyocardial biopsy**; treat the cause + HF.

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3. High-Yield Comparison Tables

Table A: Echo Differentiation (Slides 12, 18, 24) ⭐⭐

	DCM	HCM	RCM
LV cavity	**↑↑**	**↓**	**Normal**
EF	**↓↓** (systolic)	**Normal/↑** (diastolic)	**Normal** (diastolic)
Wall thickness	Normal/↓	**↑↑ (≥ 15 mm)**	Normal (*[added]* ↑ if infiltrative)
Atria	Possibly dilated	± LA enlargement	⭐ **Biatrial enlargement**
Hypertrophy	**Eccentric (series)**	**Concentric (parallel)**	—
Gallop	**S3**	**S4**	S3/S4
HF type	**HFrEF**	HFpEF	HFpEF

Table B: The Three at a Glance

	DCM	HCM	RCM
Frequency	**Commonest** (6/100,000)	1 in 500	**Rarest** (2–5%)
Genetics	40% of idiopathic are familial	**AD** (MYBPC3, MYH7)	Genetic or acquired
Classic causes	**ABCCCDD**, peripartum, tachycardia, myocarditis	Sarcomere mutations	**PLEASe Help**
Hallmark	Big floppy heart, ↓ EF	**SCD in young athletes**, SAM, LVOT obstruction	Stiff heart, big atria, **right-sided congestion**
Key exam sign	S3, displaced apex	**Murmur ↑ Valsalva, S4, bisferiens**	↑ JVP, Kussmaul's, HSM, ascites
Key test	Echo (+ exclude CAD)	Echo ≥ 15 mm + **genetics**	**CMR, endomyocardial biopsy**
Treatment	**HFrEF GDMT**, ICD/CRT	**β-blocker, verapamil**, avoid exertion, myectomy, **ICD**	**Treat the cause**, diuretics

Table C: HOCM vs Aortic Stenosis *[added]*

	HOCM	AS
Age	Young	Elderly (calcific) / bicuspid
Murmur site	Left sternal edge, **no carotid radiation**	Right 2nd ICS → **carotids**
Valsalva/standing	**↑**	↓
Squatting/handgrip	**↓**	↑ / ↓
Pulse	**Bisferiens / jerky**	**Slow-rising (parvus et tardus)**
Echo	Septal hypertrophy, **SAM**	Calcified valve, small orifice

Table D: RCM vs Constrictive Pericarditis *[added]*

	RCM	Constrictive pericarditis
Pericardium	Normal	**Thick/calcified** (CT/MRI)
Ventricular interdependence	No	**Yes** (respiratory septal bounce)
BNP	**High**	Normal/mildly ↑
Biopsy	**Infiltration/fibrosis**	Normal myocardium

	RCM	Constrictive pericarditis
Treatment	Treat the cause	**Pericardiectomy** (curable)

4. Rules, Exceptions, Patterns

- **Cardiomyopathy = intrinsic muscle disease**, not from pressure or volume overload or CAD.
- **DCM is the commonest; RCM the rarest.**
- **Series = eccentric = DCM (dilated); parallel = concentric = HCM (thick).**
- **ABCCCDD** (DCM) · **PLEASe Help** (RCM).
- **Tachycardia-induced DCM is reversible**; so is alcoholic DCM (abstinence) and often peripartum CM.
- **HCM = AD, 1/500, SCD in young athletes**; **β-blocker / verapamil**; avoid strenuous exercise.
- **HOCM murmur ↑ with ↓ preload** (Valsalva, standing, nitrates); **↓ with squatting/handgrip**.
- **Pulsus bisferiens** in HOCM (also severe AR / mixed AR+AS *[added]*).
- **Echo ≥ 15 mm wall** = HCM; **screen first-degree relatives** (echo + genetics).
- **RCM:** normal-sized ventricles, **big atria**, diastolic dysfunction; biopsy diagnoses.
- **ICD decisions:** DCM via EF ≤ 35% after GDMT (HF lecture); HCM via **risk factors**.

5. Clinical Correlations

- **Young athlete collapses on the field** → HCM (or ARVC, long QT, commotio cordis *[added]*); pre-participation ECG screening.
- **Alcohol use disorder** → DCM (reversible) and **wet beriberi** (thiamine; Substance Use lecture).
- **Chemotherapy (doxorubicin)** → baseline and serial echo (oncology).
- **Pregnancy, last month to 5 months postpartum** → peripartum CM; *[added]* bromocriptine is sometimes used; avoid ACEi in pregnancy.
- **Coxsackie B myocarditis** → DCM; **Chagas** (Latin America): apical aneurysm, conduction disease, megacolon.
- **Haemochromatosis** (bronze diabetes) → RCM/DCM; phlebotomy.
- **Sarcoidosis** → heart block, VT, RCM/DCM; CMR/PET.
- **Amyloidosis** → RCM with low-voltage ECG; also nephrotic syndrome, neuropathy, carpal tunnel.
- **Persistent fast AF** → tachycardia-mediated DCM (lecture case) → rate/rhythm control can restore EF.

6. Rapid Revision (Last-Day)

1. **Cardiomyopathy** = heart-muscle disease without pressure or volume overload or CAD. Types: **dilated, hypertrophic, restrictive**.
2. **DCM:**
   - **commonest** (6/100,000/year), **M:F 3:1**
   - ↓ contractility → **eccentric** hypertrophy (series) → dilation → ↓ EF → HF
3. **DCM causes:**
   - idiopathic 50% (40% familial)
   - myocarditis 10%
   - **ABCCCDD** (alcohol, beriberi, cocaine, Coxsackie B, Chagas, doxorubicin/daunorubicin)
   - peripartum, sarcoid, radiation, **chronic tachycardia**, endocrinopathies, HIV

4. **DCM features:** exertional dyspnoea, oedema/ascites, angina, palpitations; **S3, displaced apex**, JVD, rales.
5. **DCM tests:** ↑ BNP, troponin/CK-MB, **echo (big LV, ↓ EF, normal/thin walls)**, CXR, ECG (nonspecific). **Rx:** treat the cause + HF.
6. **HCM:**
   - **AD** (MYBPC3, β-MHC), **1 in 500**, **SCD in young athletes**
   - **concentric** hypertrophy (parallel) → ↓ relaxation → ↓ filling
   - **HOCM:** LVOT obstruction + **MR from SAM**
7. **HCM symptoms:** often none; HF, angina, **syncope**, palpitations, **SCD with exertion**.
8. **HCM signs:** crescendo–decrescendo systolic murmur, **S4**, **pulsus bisferiens**.
9. **HCM tests:** ECG (LVH/septal), **echo wall ≥ 15 mm** (small cavity, normal EF), MRI selectively, **genetic testing + family screening**.
10. **HCM Rx: avoid strenuous exercise**, **β-blockers**, **non-DHP CCB**, surgery if severe, **ICD by risk factors**.
11. **RCM: rarest (2–5%)**; **non-dilated** ventricles with impaired filling; **HF in > 80%**.
12. **RCM causes (PLEASe Help): P**ost-radiation/surgery, **L**öffler, **E**ndocardial fibroelastosis, **A**myloid, **S**arcoid, **H**aemochromatosis.
13. **RCM pathophysiology:** infiltration/fibrosis → ↓ compliance → ↑ filling pressures → **↑ atria** → pulmonary + systemic congestion.
14. **RCM features:** dyspnoea, fatigue, palpitations, **HSM, ascites**, underlying disease.
15. **RCM tests:** echo (normal cavity, EF, walls), **CMR**, CXR, **endomyocardial biopsy**. **Rx:** treat the cause + HF.
16. **Echo key:** - **DCM:** cavity ↑↑, EF ↓↓ - **HCM:** cavity ↓, walls ↑↑ - **RCM:** all ~normal + big atria

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7. Exam-Style Questions

Q1 (lecture case, slides 4–5, 26). A 48-year-old woman (father died of amyloidosis) has 2 months of exertional dyspnoea. HR 124, BP 98/60, basal crackles; ECG **AF**; CXR globular heart and pulmonary oedema; echo **dilated LV, EF 40%**. Most likely cause? a. Amyloid deposition · b. Acute psychological stress · c. Thiamine deficiency · **d. Chronic tachycardia** · e. Coronary artery occlusion

> **Answer: d** *[reasoning added; key not on slides]*.
> - **Dilated** LV with ↓ EF and **persistent fast AF** with no other cause = **tachycardia-induced DCM** (slide 9).
> - **Amyloid** causes **restrictive** CM with a non-dilated, thick-walled LV; the family history is a distractor.
> - **Takotsubo** is acute and trigger-related; **beriberi** needs malnutrition or alcohol; CAD would show regional wall-motion loss.

Q2. A 17-year-old footballer collapses during a match and is resuscitated. His father died suddenly at 35. A harsh systolic murmur at the left sternal edge **increases on standing** and **decreases on squatting**. Echo: septum 22 mm, SAM of the mitral valve. Diagnosis and inheritance? A. Aortic stenosis; sporadic · **B. Hypertrophic (obstructive) cardiomyopathy; autosomal dominant** · C. DCM; X-linked · D. ARVC; recessive

> **Answer: B** (slides 13–17).
> - **HCM:** AD sarcomere mutation (MYBPC3/MYH7).
> - **SCD in young athletes**; the murmur rises with ↓ preload.
> - He needs an **ICD** (survived arrest) and **family screening**.

Q3. In the patient above, which drug should be **avoided**? A. Bisoprolol · B. Verapamil · **C. Nitroglycerin** · D. Disopyramide

> **Answer: C** *[added; principle from slides 15, 19]*.
> - Nitrates (and diuretics, vasodilators, digoxin) **↓ preload/afterload → worsen LVOT obstruction**.
> - β-blockers and verapamil are first line (slide 19).

Q4. A 60-year-old man has dyspnoea, ascites, hepatomegaly and **raised JVP rising on inspiration**. ECG: **low voltage**; echo: **thick walls, biatrial enlargement, normal EF, small LV cavity**; nephrotic-range proteinuria. Diagnosis and best confirmatory test? A. HCM; genetic testing · **B. Restrictive cardiomyopathy (amyloidosis); biopsy (endomyocardial or fat pad)** · C. Constrictive pericarditis; pericardiectomy · D. DCM; coronary angiography

> **Answer: B** (slides 20–24).
> - **Low voltage despite thick walls** = infiltrative amyloid.
> - **Biatrial enlargement**, diastolic failure, right-sided congestion.
> - **Endomyocardial biopsy** (slide 24) or *[added]* fat-pad biopsy with **Congo red: apple-green birefringence**.

Q5. A 28-year-old woman develops dyspnoea and oedema 3 weeks after delivery; echo shows a dilated LV with EF 30%. Most likely? A. HCM · **B. Peripartum cardiomyopathy (DCM)** · C. Restrictive CM · D. Takotsubo

> **Answer: B** (slide 9 lists peripartum CM among the secondary DCM causes; *[added]* window = last month of pregnancy to 5 months postpartum).
> - Treat as HFrEF (avoid ACEi/ARNI/MRA while pregnant; they're usable postpartum, with caution if breastfeeding).
> - Anticoagulate if EF is very low.

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Cardiomyopathy from long-standing HTN or CAD	❌ By definition **not** due to pressure/volume overload or CAD (that's hypertensive/ischaemic heart disease)
HCM = commonest cardiomyopathy	❌ **DCM** is the commonest; RCM the rarest
Eccentric hypertrophy = thick walls	❌ Eccentric (series) = **dilated** (DCM); concentric (parallel) = thick (HCM)
Family history of amyloid → amyloid CM despite a dilated LV	❌ Amyloid = **restrictive**, non-dilated; dilated + AF = **tachycardia-induced DCM**

Trap	Correct thinking
HOCM murmur louder with squatting	❌ **Softer** with squatting/handgrip; **louder** with Valsalva/standing
Nitrates or diuretics for HOCM angina/dyspnoea	❌ Worsen obstruction; **β-blocker/verapamil**
Digoxin in HCM or amyloid *[added]*	❌ Avoid (↑ obstruction; toxicity in amyloid)
HCM echo threshold 10 mm	❌ **≥ 15 mm** (*[added]* ≥ 13 mm in relatives)
HCM has a low EF	❌ EF **normal or high**; diastolic dysfunction
RCM echo always shows normal walls	⚠️ Infiltrative causes (amyloid) give **thick walls**; slide says "typically normal"
RCM = constrictive pericarditis	❌ Different: pericardial thickening, septal bounce, BNP normal in constriction; biopsy abnormal in RCM
ICD for every HCM patient	❌ Guided by **risk factors** (prior arrest, FHx SCD, syncope, wall ≥ 30 mm, NSVT…)
Only the patient needs evaluation in HCM	❌ **Screen first-degree relatives** (AD: 50% risk)
Peripartum CM only during pregnancy	⚠️ Last month of pregnancy up to **5 months postpartum** *[added]*